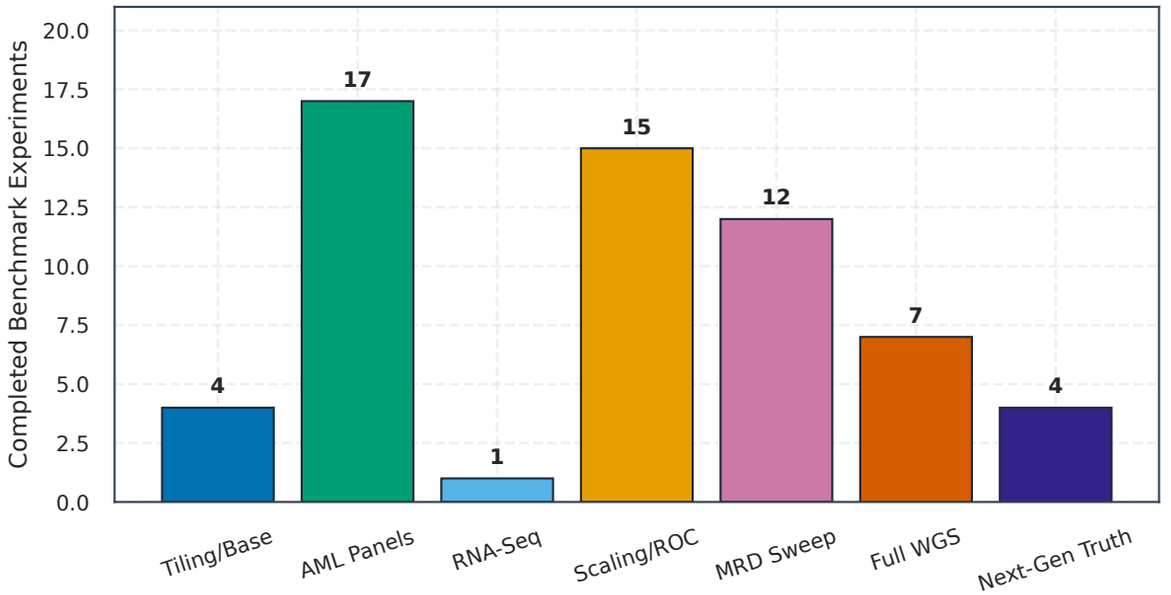
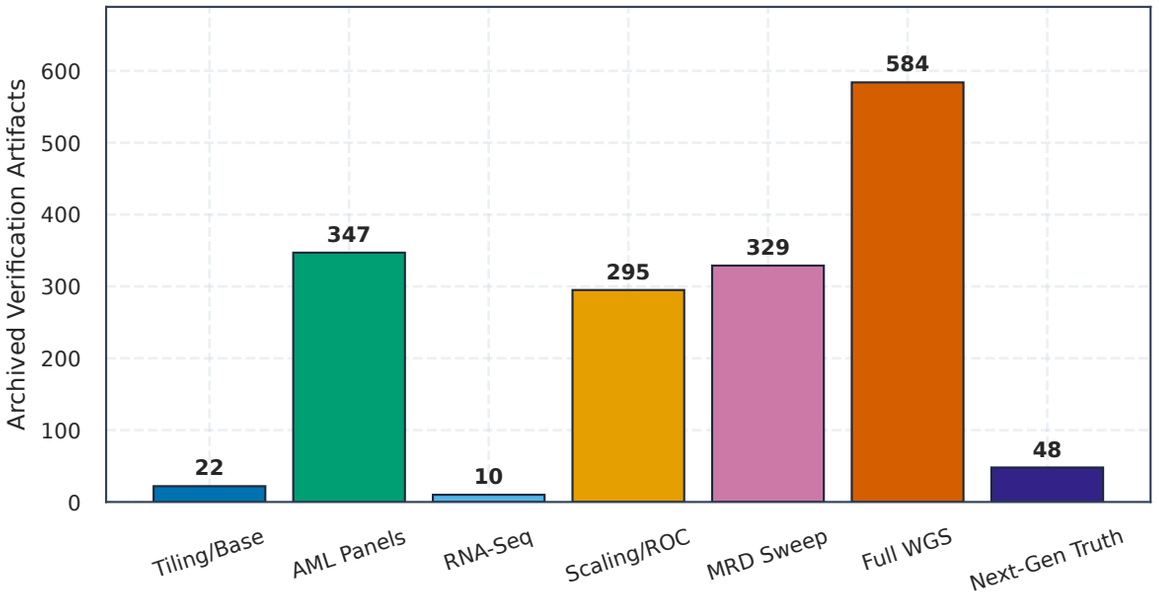
A. Multi-Modal Empirical Evaluation Architecture Across Six Benchmark Suites (VarDictRust v1.0.6 vs. VarDictJava v1.8.3 & vardictcpp)

1. Tile-Size Invariance &	2. Clinical AML Panels &	3. Transcriptome	4. Multi-Core Exome	5. Multi-Regime (.var)	6. Full-Genome WGS	7. Next-Gen Truth Sets
Multi-Region Parity (0.5–10 Mbp Tiles)	13-Mode Functional Equivalence Suite	Splice-Aware RNA-Seq (51.8 GiB HG002)	Scaling & Chr1 Span Stress	& Ultra-Deep MRD Parity Sweep	& 10-Chemistry Platform Panel	(T2T-Q100 v5.0q, CMRG, Mosaic & HG008 v0.4)
4 experiments | 22 artifacts **100% Completed**	**17 experiments | 347 artifacts** **100% Completed**	**1 experiments | 10 artifacts** **100% Completed**	**15 experiments | 295 artifacts** **100% Completed**	**12 experiments | 329 artifacts** **100% Completed**	**7 experiments | 584 artifacts** **100% Completed**	**4 experiments | 48 artifacts** **100% Completed**

Hardware Lock: GCP n2-standard-32 (32 vCPU, 128 GiB RAM, NVMe RAID0) | Comparators: VarDictJava v1.8.3 & vardictcpp | 100% Reproducible

B. Benchmark Experiments per Evaluation Suite



C. Archived Verification Artifacts per Suite